

Discussion 7: Solutions

Consider two continuous random variables X and Y with joint p.d.f.

$$f_{X,Y}(x,y) = \begin{cases} C(x+2y) & 0 < x < 2, 0 < y < 3 \\ 0 & \text{otherwise} \end{cases}$$

(1-1) Find the value of C so that $f_{X,Y}(x,y)$ is a valid joint p.d.f.

(1-2) Find the marginal probability density function of X , $f_X(x)$.

(1-3) Find the marginal probability density function of Y , $f_Y(y)$.

(1-4) * Are X and Y independent? If X and Y are not independent, find $Cov(X,Y)$.

[(1-1)]

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx dy = C \int_0^3 \int_0^2 (x+2y) dx dy \\ 1 &= C \int_0^3 \left(\frac{x^2}{2} + 2yx \right) \Big|_0^2 dy \\ 1 &= C \int_0^3 (2 + 4y) dy \\ 1 &= C(2y + 2y^2) \Big|_0^3 \\ 1 &= 24C \\ C &= \frac{1}{24} \end{aligned}$$

[(1-2)] When $x < 0$ or $x > 2$, $f_{X,Y}(x,y) = 0$. When $0 < x < 2$,

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x,y) dy = \frac{1}{24} \int_0^3 (x+2y) dy \\ f_X(x) &= \frac{1}{24} (xy + y^2) \Big|_0^3 = \frac{1}{24} (3x + 9) = \frac{x+3}{8} \end{aligned}$$

Hence,

$$f_X(x) = \begin{cases} \frac{x+3}{8} & 0 < x < 2, \\ 0 & \text{otherwise} \end{cases}$$

[(1-3)] When $y < 0$ or $y > 3$, $f_{X,Y}(x, y) = 0$. When $0 < y < 3$,

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \frac{1}{24} \int_0^2 (x + 2y) dx$$

$$f_Y(y) = \frac{1}{24} \left(\frac{x^2}{2} + 2yx \right) \Big|_0^2 = \frac{1}{24} (2 + 4y) = \frac{1 + 2y}{12}$$

Hence,

$$f_Y(y) = \begin{cases} \frac{1 + 2y}{12} & 0 < y < 3, \\ 0 & \text{otherwise} \end{cases}$$

[(1-4)]* X and Y are not independent, since

$$f_{X,Y}(x, y) \neq f_Y(y)f_X(x)$$

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^2 x \frac{x+3}{8} dx = \int_0^2 \frac{x^2 + 3x}{8} dx = \frac{1}{8} \left(\frac{x^3}{3} + \frac{3x^2}{2} \right) \Big|_0^2 = \frac{13}{12}$$

$$E[Y] = \int_{-\infty}^{\infty} y f_Y(y) dy = \int_0^3 y \frac{1+2y}{12} dy = \int_0^3 \frac{y + 2y^2}{12} dy = \frac{1}{12} \left(\frac{y^2}{2} + \frac{2y^3}{3} \right) \Big|_0^3 = \frac{15}{8}$$

$$E[XY] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x, y) dx dy = \int_0^3 \int_0^2 \frac{xy(x+2y)}{24} dx dy = \int_0^3 \frac{y}{24} \int_0^2 (x^2 + 2yx) dx dy$$

$$E[XY] = \int_0^3 \frac{y}{24} \left(\frac{x^3}{3} + yx^2 \right) \Big|_0^2 dx dy = \int_0^3 \frac{1}{6} \left(\frac{2}{3}y + y^2 \right) dy = \frac{1}{6} \left(\frac{y^2}{3} + \frac{y^3}{3} \right) \Big|_0^3 = 2$$

$$Cov(X, Y) = E[XY] - E[X]E[Y] = 2 - \left(\frac{15}{8} \right) \left(\frac{13}{12} \right) = 2 - \frac{65}{32} = \frac{64 - 65}{32} = -\frac{1}{32}$$

2. A random variable X has a Exponential distribution with mean 0.2. Find $P(\mu - \sigma < X < \mu + \sigma)$.

$X \sim Exp(\lambda)$. Then $E[X] = \mu = \frac{1}{\lambda} = 0.2$, $\sigma = \frac{1}{\lambda} = 0.2$, $\lambda = 5$.

$$P(0 < X < 0.4) = \int_0^{0.4} 5e^{-5x} dx = -e^{-5x} \Big|_0^{0.4} = 1 - e^{-2}$$

3. 60 scientists from 30 universities attend a conference. The conference includes a lunch in a cafeteria which has 30 tables each suitable for 2 people. The people seated for lunch at random. Let $X_j = 1$ if the scientists from university j seat together and $X_j = 0$ otherwise.

- (a) Compute $E(X_1)$ and $V(X_1)$.
- (b) Compute $Cov(X_1, X_2)$.
- (c) Let $X = X_1 + X_2 + \cdots + X_{30}$ be the number of tables occupied by the people from the same university. Compute $E[X]$ and $V(X)$.

$$X_j = \begin{cases} 1 & \text{if Scientists from university } j \text{ seat together} \\ 0 & \text{otherwise} \end{cases}$$

$$P(\text{Seating together}) = \frac{60 \cdot 1}{60 \cdot 59} = \frac{1}{59}$$

$$X_j \sim \text{Bern}(p) \text{ where } p = P(X_j = 1) = P(\text{Seating together}) = \frac{1}{59}$$

$$E[X_j] = \frac{1}{59}, \text{Var}(X_j) = \frac{1}{59}(1 - \frac{1}{59}) = \frac{58}{59^2}$$

$$Cov(X_1, X_2) = E[X_1 \cdot X_2] - E[X_1]E[X_2]$$

$$E[X_1 \cdot X_2] = P(X_1 = 1 \text{ and } X_2 = 1) = P(X_2 = 1 | X_1 = 1)P(X_1 = 1) = \frac{58 \cdot 1}{58 \cdot 57} \cdot \frac{60 \cdot 1}{60 \cdot 59} = \frac{1}{57} \cdot \frac{1}{59}$$

$$Cov(X_1, X_2) = \frac{1}{57} \cdot \frac{1}{59} - \frac{1}{59} \cdot \frac{1}{59} = \frac{2}{(59)^2 57}$$

$$\text{Generally } Cov(X_i, X_j) = \frac{2}{(59)^2 57} \text{ when } i \neq j.$$

$$\begin{aligned} Cov(\sum_{i=1}^{30} X_i, \sum_{i=1}^{30} X_j) &= \sum_{i=1}^{30} Var(X_i) + 30 \cdot 29 Cov(X_1, X_2) \\ &= 30 Var(X_1) + 30 \cdot 29 Cov(X_1, X_2) \\ &= 30 \frac{58}{59^2} + 30 \cdot 29 \frac{2}{(59)^2 57} \\ &= \frac{30}{57} \left(\frac{58}{59} \right)^2 \end{aligned}$$

4*. The joint density function of X and Y is given by

$$f(x, y) = \begin{cases} \frac{15}{2}x(2 - x - y) & 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Compute the conditional density of X , given that $Y = y$, where $0 < y < 1$.

$$f(X = x|0 < y < 1) = \frac{f(X = x, Y = y) \text{ and } 0 < y < 1}{f_Y(y) \text{ when } 0 < y < 1} =$$

$$\int_{-\infty}^{\infty} f(x, y) dx = \frac{15}{2} \int_0^1 (2x - x^2 - yx) dx = \frac{15}{2} \left(x^2 - \frac{x^3}{3} - \frac{yx^2}{2} \right) \bigg|_0^1 = \frac{5}{4}(4 - 3y)$$

$$f_Y(y) = \begin{cases} \frac{5}{4}(4 - 3y) & 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

if $0 < x < 1$, then

$$f(X = x|0 < y < 1) = \frac{f(X = x, Y = y) \text{ and } 0 < y < 1}{f_Y(y) \text{ when } 0 < y < 1} = \frac{\frac{15}{2}x(2 - x - y)}{\frac{5}{4}(4 - 3y)} = \frac{6x(2 - x - y)}{4 - 3y}$$

$$f(X = x|0 < y < 1) = \begin{cases} \frac{6x(2 - x - y)}{4 - 3y} & 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

5.

$x = 1$	$\frac{1}{6}$	$\frac{1}{8}$	$\frac{1}{4}$
$x = 0$	$\frac{1}{8}$	$\frac{1}{12}$	$\frac{1}{4}$
	$y = 0$	$y = 1$	$y = 2$

(a) Find $P(X = 0|Y \leq 1)$.

(b) Find the marginal PMFs of X and Y .

(c) Find $P(Y = 1|X = 0)$.

(d) Are X and Y independent?

$$(a) P(X = 0|Y \leq 1) = \frac{P(X = 0, Y = 1) + P(X = 0, Y = 0)}{P(Y = 0) + P(Y = 1)} = \frac{\frac{1}{12} + \frac{1}{8}}{\frac{1}{12} + \frac{1}{8} + \frac{1}{6} + \frac{1}{8}} = \frac{\frac{2+3}{24}}{\frac{2+3+4+3}{24}} = \frac{5}{12}.$$

$$\begin{aligned}
 \text{(b) } P(X=0) &= \frac{1}{8} + \frac{1}{12} + \frac{1}{4} = \frac{11}{24} & P(X=1) &= \frac{1}{6} + \frac{1}{8} + \frac{1}{4} = \frac{13}{24} \\
 P(Y=0) &= \frac{1}{8} + \frac{1}{6} = \frac{7}{24}, & P(Y=1) &= \frac{1}{8} + \frac{1}{12} = \frac{5}{24} & P(Y=2) &= \frac{1}{2}.
 \end{aligned}$$

$$\text{(c) } P(Y=1|X=0) = \frac{P(X=0, Y=1)}{P(X=0)} = \frac{\frac{1}{12}}{\frac{11}{24}} = \frac{2}{11}$$

$$\text{(d) } X \text{ and } Y \text{ are not independent, for example } P(X=0)P(Y=0) \neq P(X=0, Y=0)$$